Low Cost Sensor With IoT LoRaWAN Connectivity and Machine Learning-Based Calibration for Air Pollution Monitoring

Sharafat Ali, *Graduate Student Member, IEEE*, Tyrel Glass, Baden Parr, Johan Potgieter, and Fakhrul Alam, *Senior Member, IEEE*

***Abstract*— Air pollution poses significant risk to environment and health. Air quality monitoring stations are often confined to a small number of locations due to the high cost of the monitoring equipment. They provide a low fidelity picture of the air quality in the city; local variations are overlooked. However, recent developments in low-cost sensor technology and wireless communication systems like Internet of Things (IoT) provide an opportunity to use arrayed sensor networks to measure air pollution, in real time, at a large number of locations. This article reports the development of a novel low-cost sensor node that utilizes cost-effective electrochemical sensors to measure carbon monoxide (CO) and nitrogen dioxide (NO_2) concentrations and an infrared sensor to measure particulate matter (PM) levels. The node can be powered by either solar-recharged battery or mains supply. It is capable of long-range, low power communication over public or private long-range wide area network (LoRaWAN) IoT network and short-range high data rate communication over Wi-Fi. The developed sensor nodes were co-located with an accurate reference CO sensor for field calibration. The low-cost sensors' data, with offset and gain calibration, show good correlation with the data collected from the reference sensor. Multiple linear regression (MLR)-based temperature and humidity correction results in mean absolute percentage error (MAPE) of 48.71% and R^2 of 0.607 relative to the reference sensor's data. Artificial neural network (ANN)-based calibration shows the potential for significant further improvement with MAPE of 38.89% and R^2 of 0.78 for leave-one-out cross-validation.**

***Index Terms*— Air pollution monitoring, air quality monitor (AQM), Internet of Things (IoT), long-range wide area network (LoRaWAN), low-cost sensor, machine learning, remote sensing, sensor calibration.**

I. Introduction

AIR pollution has significant impact on the quality of life and public health [1]. Monitoring of air pollution is of considerable importance for raising public awareness and improvement in human health and sustainable urban environments. While air quality monitoring stations are commonly deployed across cities to monitor pollutants, they are usually spread across wide areas due to the cost of accurate sensors. As an alternative, a large network of sensor nodes consisting of inexpensive instrumentation are being proposed to supplement the conventional system [2]. Researchers have started to develop low-cost sensors and other relevant supporting technologies that can be used to gather real-time air pollution data [3]. Many innovative methods have been proposed to improve the accuracy and the operability of the low-cost sensors [4]. While these low-cost sensors are not as accurate or reliable as their conventional (and expensive) counterparts [5], a key benefit of this approach is that spatial patterns in air pollutant concentrations can be presented visually in near real time. The lack of precision in sensor measurements, which can be improved, is offset by the ability to collect large amounts of data at a relatively low cost. Statistical techniques can be used to analyze temporal and spatial patterns in the data in multiple ways to investigate both the impacts of local pollutant sources and long-range pollutant drift [6].

The five major pollutants typically reported to describe air quality are, namely, sulfur dioxide (SO_2), carbon monoxide (CO), ground-level ozone (O_3), nitrogen dioxide (NO_2), and suspended particulate matter (PM) [7]. Air quality index (AQI) is often defined based on the concentration of these pollutants.

Environmental parameters and air quality monitoring based on wireless sensor networks have been extensively reported in the literature (see [8], [9]). The common communications protocols used in these networks have been a combination of cellular [10] and short-range wireless network technology (e.g., Bluetooth, ZigBee, etc.) [11], [12]. Recent development in Internet of Things (IoT) can facilitate the deployment of sensors over wide areas and in remote locations. However, some of the reported works still require the support of traditional telecommunication infrastructure (e.g., cellular backhaul, Wi-Fi gateway, etc.) for remote operation [13]–[16]. A system to monitor air pollution with high spatio-temporal resolution requires a network architecture that is capable of providing a large coverage with comparatively low power consumption. Low power wide area (LPWA) technologies may provide a suitable solution [17]. Many forms of LPWA technologies are available, e.g., narrowband IoT (NB-IoT), long term evolution for machines (LTE-M), SigFox, and

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long-range (LoRA) wide area network (WAN) or LoRaWAN [18]. LPWA is becoming popular for many applications like water quality monitoring [19], asset monitoring [20], and a few studies on LPWA-based air pollution studies have been reported in the literature.

Duangsuwan *et al.* [21], [22] report a sensing system for monitoring multiple air pollutants in the urban environment using NB-IoT architecture. However, they do not report any calibration or validation. Tzortzakis *et al.* [23] present a self-powered system based on LoRa for air pollution monitoring. The sensors still rely on a cellular gateway for connecting to the cloud. They also do not report any calibration of the low-cost sensors. Rosmiati *et al.* [24] develop a LoRa-based real-time ambient air monitoring system using Raspberry-Pi3 to manage data transmission. The developed prototype requires complex and costly system architecture and therefore may not be suitable for large-scale deployment. Rossi and Tosato [25] propose an IoT system for pollution monitoring with low-cost gas and PM sensors using LoRa communication link. However, their results are not validated for field deployments. Sujuan *et al.* [26] develop a sensor node for monitoring multiple pollutants with LoRa communication. However, the developed system appears to be not scalable enough to cope with large-scale IoT-based deployment and did not have any data validation. Hsieh *et al.* [27] develop a vehicular sensor system using LoRa communication link for measuring multiple pollutants. But their approach seems unsuited for large city-scale deployment. They also do not perform any rigorous sensor calibration. Thu *et al.* [28] report the development of a LoRa-enabled sensor node to monitor PM and carbon dioxide (CO_2) along with temperature and humidity. While they discuss how statistical models can be used for temperature and humidity forecasting, they do not perform any calibration of the low-cost pollutant sensors. Filip *et al.* [29] report the development of a LoRa-based sensor node to monitor multiple pollutants. Unfortunately, it seems to be a work in progress and lacks details about data collection, calibration, etc. Also, the nodes appear to be powered by mains supply and therefore are not suitable for remote, autonomous operation in current form. LoRaWAN-based systems for air pollution monitoring have also been reported in [30]–[32]. However, these are designed to monitor only one pollutant, the PM.

This article extends the work presented in [33] and reports the development of a novel IoT-enabled low-cost sensor for monitoring air pollution. The custom-designed sensor node has the flexibility of using both LoRaWAN IoT network and Wi-Fi for long-range and short-range communication, respectively. The sensor nodes can be manufactured from parts that cost approximately U.S. $250. This article also discusses the mechanical design and network configuration of the sensor. The developed sensor nodes are deployed for testing and data collection. The CO sensors are calibrated in a field setup by comparing their readings with a collocated accurate, reference sensor. Neural network (NN)-based multivariate calibration shows the potential to improve the accuracy of the sensor significantly. The developed sensors may facilitate a city-wide distributed sensor deployment for high-density geospatial monitoring of multiple air pollutant in near real time.

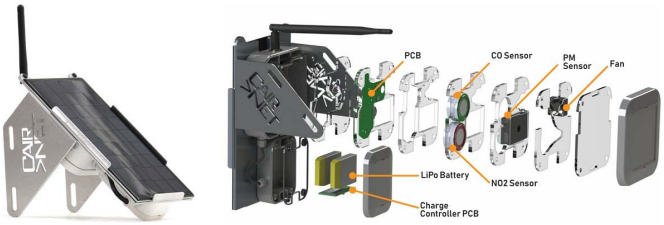


Fig. 1. Developed low-cost sensor node. Sensor and communication modules are housed in one enclosure with power management in the second enclosure.



Fig. 2. Printed circuit board (PCB) details of the sensor and wireless communication module. Quad Opamp is the major component of the driver circuit (see Fig. 3).

II. Overall Design

The developed sensor unit is solar-powered, weather-resistant, and designed to be easily mounted to lamp posts or walls and capable of measuring CO, NO_2, PM (PM_{10}) along with temperature and humidity. Fig. 1 shows the prototype sensor module.

Two of the most widely reported low-cost gas sensors to measure the ambient air pollution are metal oxide (MOX) sensors [34] and electrochemical (EC) sensors [35]. Although MOX sensors are small in size and have wide commercial availability at low cost [36], EC sensors have been recognized as better suited for real-time ambient air quality monitoring [2], [3], [35]. Previous researches also show that EC sensors used to determine CO and NO_2 concentrations may be suitable for urban air quality measurements as they respond in a highly linear fashion to the concentration of their respective target gases [37]. The design is flexible enough to swap the gas sensors with O_3 and SO_2 sensors with appropriate driver circuitry. EC gas sensors (Alphasense CO-B4 [38] and NO2-B43F [39] for CO and NO_2, respectively) were selected for the sensor design based on their promising results reported in [40].

Most widely used low-cost sensors for measuring PM are based on optical sensing or light scattering [35]. These sensors have demonstrated quick response time while having low manufacturing cost and lower power requirements [41]. The Sharp optical sensor GP2Y1010AU0F [42] was selected for its low price and comparable performance in chamber tests to other popular optical scattering units [41]. It was also found to perform well in real-world deployments [43].

The sensor node consists of three main systems: sensors, wireless communication, and power management (please see Figs. 1 and 2). The sensor module consists of the sensors for NO_2, CO, PM, temperature, and humidity as well as the necessary circuitry to drive these sensors. The temperature and humidity sensors are necessary as the gas sensors have dependence on these ambient parameters and thus need to be compensated as outlined by the manufacturer [44].

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TABLE I
COMPONENT COST TO BUILD THE SENSOR

Component	Approximate Cost (USD)
Enclosure Cases	$9.63
LiPo Battery	$10.71
MCU/LoRa Radio	$14.59
Solar Panel	$20.96
Alphasense NO_2 Sensor	$59.32
Alphhasense CO Sensor	$57.94
Sharp PM Sensor	$14.28
Miscellaneous	$62.42
Total Cost:	$249.86

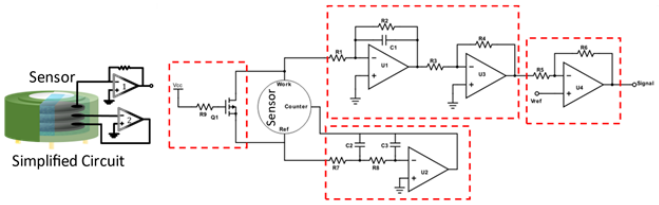


Fig. 3. Driver circuit. An elementary representation is shown on the left. The detailed circuit is on the right.

The communications module utilizes a chipset that contains a microcontroller, which receives and stores data, and a radio, which transmits data over the IoT network. The power management module utilizes a solar panel and custom-designed circuitry to recharge the LiPo batteries that power all the electronics components of the node. This module also hosts an ESP8266 Wi-Fi module [45] to add communications flexibility. The Wi-Fi module internally connects to the communication module via universal asynchronous receiver/transmitter (UART).

Table I shows the breakdown of cost for the parts required to manufacture the sensor node. Excluding the cost of PCB manufacturing and assembling, one sensor node can be constructed for approximately U.S. $250. This is significantly cheaper compared to commercially available air pollution sensors. A handheld sensor that measures multiple pollutants can easily cost U.S. $5000.[1] Sensors used for permanent installations cost many times more than that. However, it should be noted that such expensive sensors are expected to be more accurate than the low-cost sensors.

A. Driver Circuitry

The EC gas sensors are EC cells that generate current in proportion to target gas concentration [46]. Oxidation/reduction of target gas needs to be balanced by a complementary reaction, which is facilitated by the driver circuitry. Its other main role is to amplify and condition the current signal so it can be read by an ADC.

The driver, as shown in Fig. 3 in its elementary configuration, consists of two major components. The first is trans-impedance amplifier for the small signal current that also clamps the working electrode to the ground. The second component ensures that the counter reaction balances the main reaction. It accomplishes this task by comparing the potential of the working electrode (ground) to the electrolyte solution using a third reference electrode that is not involved with the reaction. Fig. 3 also shows a more detailed schematic of the driver circuitry. U2 also provides some low pass filtering with the additional *RC* network. The trans-impedance amplifier is split over two stages. The first stage (U1) has minimal gain to allow the circuit to respond to larger current changes. It also performs some low pass filtering. U3 then compensates for the reduced gain. U4 is another low gain stage that offsets the signal to utilize the full dynamic range of the microcontroller. A transistor (Q1) acts as a flywheel when the circuit powers down during and clamps the working and counter electrodes together allowing the reaction to precede without charge buildup.

The gas sensors are factory precalibrated. The two parameters measured are the zero current (the current when no target gas is present) and sensitivity [46]. The voltage read at the microcontroller's ADC is mapped to the concentration of the target gas using these calibration data in conjunction with the driver circuitry biasing values.

The PM sensor comes with a built-in drive circuitry and requires a 5-V power supply, digital input to control the light, and an ADC channel to measure the output.

B. Microcontroller/Radio Unit

The sensor node utilizes a Murata CMWX1ZZABZ-078 chipset [47] consisting of a low power microcontroller paired with LoRaWAN radio. The microcontroller contains the firmware for implementing the LoRaWAN protocol stack but also provides the embedded computation for measuring the signals from the sensors. Five ADC channels are required for collecting the sensor data, two for each of the gas sensors, and one for the PM sensor. The temperature/humidity sensor uses I2C protocol to transfer data. General-purpose input–output (GPIO) pins are used to control the fan (please refer to Section II-C) and the PM sensor.

Before transmission, the data from the five sensors are grouped into a 128-bit packet and transmits the data at a career frequency of 923 MHz. The microcontroller communicates with the ESP8266 Wi-Fi chip in the battery module via UART. This provides a means of network debugging and allows for a much higher data rate than LoRaWAN. However, Wi-Fi is much more power-intensive and must have a suitable power supply (e.g., mains). The UART/Wi-Fi interface also provides an alternate means for firmware updates of the nodes.

C. Mechanical Design

The node is comprised of two main enclosures (please see Fig. 1). The first one houses the sensors along with the drivers and communication circuitry, while the second contains the batteries and solar charge controller. Splitting the functionality this way means only the first enclosure is required when a power source is available. Both cases are rated IP67, but the sensor case has had air vents installed to allow flow through the unit. The cases are mounted to an aluminum bracket that clamps on a lamppost with the solar panel on the opposite side. Fig. 4 shows the assembled sensors/communication module.

To take a new air pollutant measurement, the fan must draw air through the chamber until the whole volume has

[1] https://www.aeroqual.com/product/outdoor-portable-monitor-pro-kit

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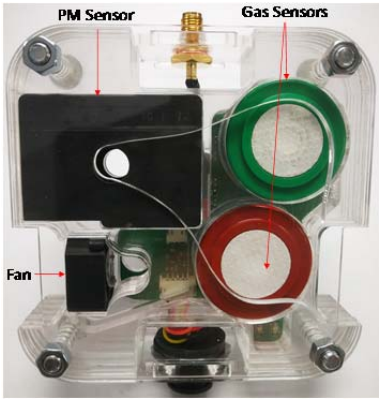


Fig. 4. Assembled sensor/communication modules.

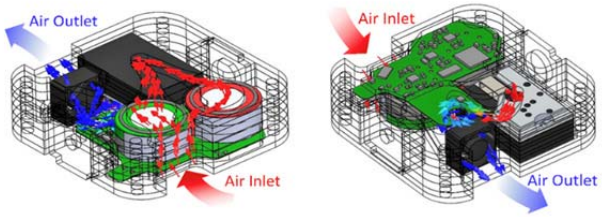


Fig. 5. Airflow simulation.

been refreshed. Air enters the cavity and is split into two streams passing over the gas sensors before recombining and entering the PM sensor. Fig. 5 shows the SOLIDWORKS flow simulation used to ensure the airstream was split evenly over the two sensor faces.

After leaving the PM sensor, the air is directed around a curved channel where it comes in direct contact with the PCB and surface mounted temperature/humidity sensor. Finally, the air stream passes through a hole in the PCB, through the fan and out of the unit. The reading takes place only once all the air has been replaced.

The fan is a 5-V single-phase eight-pole brushless dc motor with a flow rate of 707.921 cm^3/s at a current draw of 155 mA. The total enclosed air volume of the sensor is 9.2 cm^3 which at the stated flow rate would take 13 ms to replenish. However, to allow for the fan to reach operating speed and initiate airflow it is powered for 60 ms during each reading.

D. Network Configuration

The data collected from the sensor nodes are ultimately meant to be displayed on a web page for end-users (please see Fig. 6). For the data to reach an end-user, it must be transported in three different stages: from device to gateway, then from gateway to network server, and finally from network server to application server. The developed sensor nodes operate over a LoRaWAN IoT network. In addition, the data can be sent over Wi-Fi (if an access point is available) by piping out data to the ESP8266 Wi-Fi chip hosted in the power management module.

LoRaWAN allows for flexibility in the use of gateways. A proprietary gateway can be used to receive the IoT data. Alternatively, the sensors can connect to a commercial network of gateways. The advantage of this approach is the flexibility

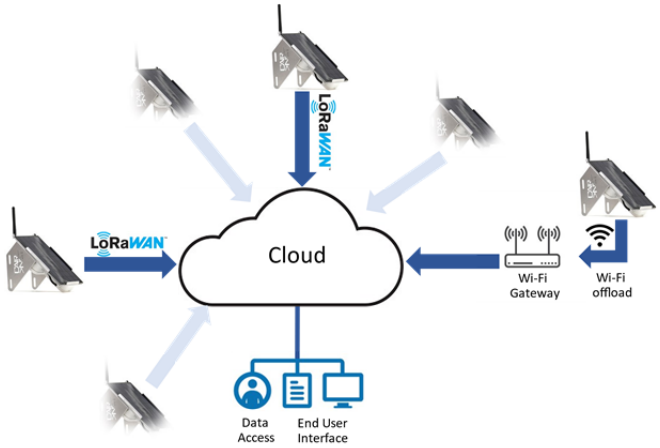


Fig. 6. Proposed air quality monitor (AQM) network.

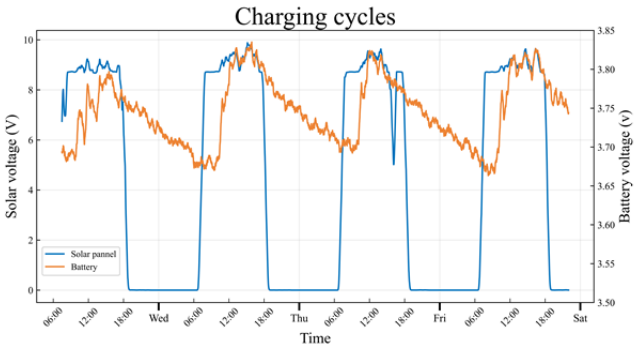


Fig. 7. Example of charge–discharge cycle of a sensor over a four-day period. The solar panel powers the sensor and charges the battery (battery voltage rising) when sufficient sunlight is available. The battery powers the sensor (battery voltage falling and zero output voltage for the solar panel) otherwise.

in deployment over a much larger area without needing to manage several gateways but per message costs apply.

Forwarding of the packets from the gateway to a desired Internet endpoint, such as an application server or a database, is similar for the two gateway options. The network server responsible for routing the packets received by the gateway is configured to send the data to specific application servers depending on which sensor nodes the data originated from.

E. Power Management

Power to the sensor node is provided by a 4000-mAh LiPo battery fed through a switch mode power supply (SMPS). A 3 W, 9-V solar panel charges the battery and provides power to the unit when adequate sunlight is available. An LT3652[2] solar battery charging IC provides a two-state constant current—constant voltage charge cycle (please see Fig. 7) along with current regulation to ensure peak solar output power. However, when the node is operating using Wi-Fi, the expectation is that it will be mains powered.

The sensor node has both fixed and variable current (i.e., power) draws. The fixed draw is required regardless of whether measurements are being taken. Background processes

[2]https://www.digikey.co.nz/en/product-highlight/l/linear-tech/lt3652-2a-battery-charger

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that include the standby power consumptions for the microcontroller, temperature and humidity sensors, and switching regulators amount to approximately 0.6 mA. The current for the gas sensor circuits at 0.556 mA is the major contributor. However, the power is budgeted to support a constant draw of 2 mA (i.e., 48 mAh over a 24-h period) to provide excess capacity for datalogging or on-board computation in the future.

Taking a new reading consumes power from several sources. Therefore, the power consumption of acquiring and transmitting new readings introduces a tradeoff between reading frequency and power conservation. Drawing fresh air through the sensor with the fan (please see Section II-C for details) incurs intermittent power consumption. While actively taking readings, the microcontroller needs to exit low power mode. The amount of time spent out of sleep mode is dependent on how long it takes to take measurements and time required to send the data packets. In the default mode, the sensor node transmits data at one-minute interval. Each minute, measurements are taken over a 3 s window with three 1-s slots reserved for temperature/humidity, PM and gas sensors (CO/NO_2) each. During the respective one-second slots, 25 temperature/humidity readings are taken whereas 100 readings are taken for the PM, CO, and NO_2 to reduce noise by averaging. Once collected, the data are sent using the LoRa protocol. The time taken for a transmission depends on the spreading factor, bandwidth, and packet size. However, many regional LoRaWAN specifications dictate a 1% maximum transmission duty cycle [48] resulting in on-air time of 600 ms for a 1-min transmission frequency. For class A operation, the radio must also be actively listening for a downlink message in the two timeslots directly following the transmission (164 ms each) and thereby consumes energy. The PM sensor is the other major source of power consumption. The temperature and humidity sensors add only a negligible amount. Table II shows a detailed breakdown of the energy consumptions at a frequency of one transmission per minute over LoRaWAN within a 24-h period. As can be seen, the total energy consumption is equivalent to approximately 94 mAh which is far smaller than the capacity of the battery. Also, as discussed previously, the solar panel charges the battery and also powers the sensors depending on sunlight availability.

The energy harvesting capabilities of the solar panel are affected by the sunlight available. The daily equivalent full-sunshine hours at the deployment location over a 5-year period[3] was used to conduct preliminary analysis on the sensor node's ability to continue remote operation without access to mains power. While the average full sunshine hours per day, 6.33, is adequate for the 3-W panel to fully charge the battery, there are days where there is no sunshine at all. Such days are often clustered and will force the sensor node to rely solely on the battery without little to no recharging opportunity over that duration. LiPo batteries suffer degradation with the charge–discharge cycles. This degradation is dependent on the depth of discharge. A big discharge depth can be quite detrimental. If the battery is assumed to be fully charged at the beginning of the historical data, for the default mode of one-minute transmit frequency, the maximum discharge depth (coinciding with clustered low sunlight days) is estimated to be less than 10%. Therefore, the sensor nodes are anticipated to maintain normal operation without degrading the battery health over a long period of time. Also, the bidirectional communication capabilities of LoRa allows for remote commands to reduce the transmission frequency during periods of abnormally low sunlight.

[3]https://cliflo.niwa.co.nz/ has such historical data available for New Zealand.

TABLE II
ENERGY CONSUMPTION OF THE SENSOR

Component	Approximate Energy Consumption (mAh) / 24 hour
PM Sensor	12.1
Fan	5.6
LoRa TX	21.6
LoRa RX	1.8
Microcontroller	4.7
T/H Sensors	0.01
Fixed Draw	48
Total Energy Consumption:	93.81

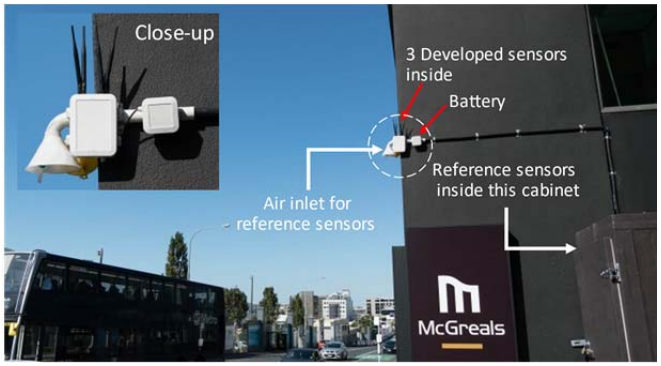


Fig. 8. Deployment for field calibration at Khyber Pass, Auckland.

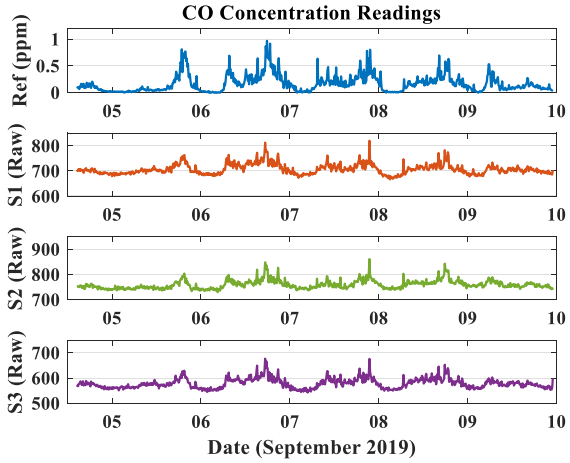


Fig. 9. Raw CO sensor data.

III. DEPLOYMENT AND FIELD CALIBRATION OF CO SENSOR

The developed sensor needs to be calibrated so that its measurements closely agree with the reference measurements from a reliable high-quality sensor [49]. Calibration models for low-cost air pollutant sensors are strategies followed to compare its raw measurement readings with that from a

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TABLE III
CALIBRATION RESULTS

Calibration Method	Sensor S1			Sensor S2			Sensor S3		
	MAPE	R^2	RMSE	MAPE	R^2	RMSE	MAPE	R^2	RMSE
OLS	57.15	0.56	0.107	61.43	0.541	0.109	60.98	0.524	0.111
MLR	48.71	0.607	0.101	52.54	0.542	0.109	49.12	0.58	0.104
ANN	38.89	0.78	0.075	40.01	0.78	0.076	38.83	0.78	0.076

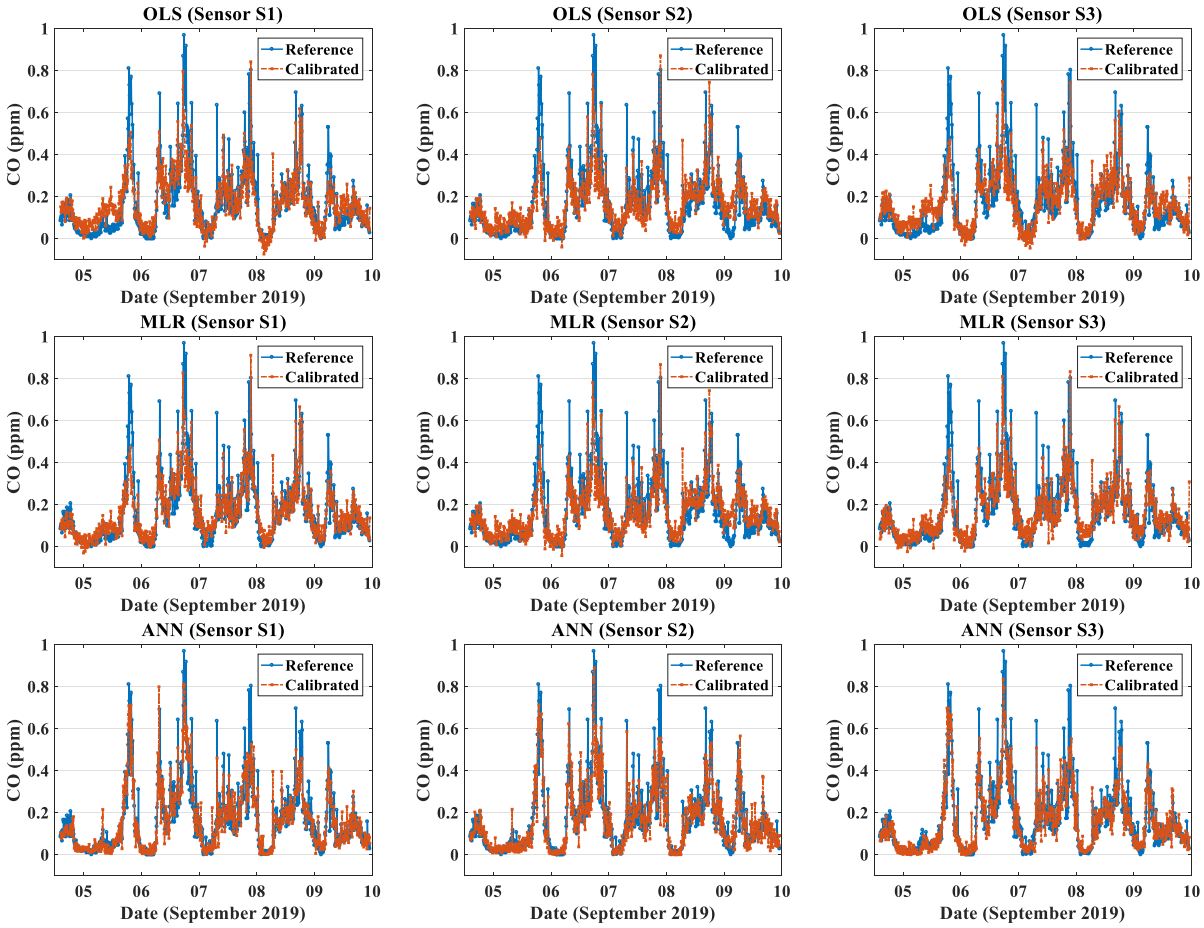


Fig. 10. Calibrated CO sensor output for different calibration methods.

reliable reference sensor and transforming them to more accurate calibrated values. The calibration setup can be classified into two general groups: laboratory setup under a controlled environment and field setup under an observed environment with the sensor co-located at reference monitoring site. The alphasense gas sensors came with an initial laboratory calibration for offset and gain. Therefore, it was decided to field calibrate the sensors by co-locating them next to a more accurate reference sensor. The reliable readings from the reference monitoring stations are used to train and evaluate. Many researchers recommend such field setups for calibration [50], as it takes into account the inherently complex nature of the ambient conditions [51]–[53].

Three sensor nodes were manufactured following the development process described in Section II. The nodes were tested extensively in the lab with regards to their connectivity both over Wi-Fi and LoRaWAN. Afterward, they were deployed for field testing. The three nodes were placed within a single enclosure and installed next to a reference AQM station maintained by the Auckland City Council. The AQM station among other instruments contains a Teledyne API Model 300-A CO [54] that is operated and maintained in accordance with the manufacturer's instructions and the respective Australian standards/New Zealand standards. Fig. 8 shows the installed sensor nodes at Khyber Pass, Auckland. The CO data along with temperature and relative humidity (RH) were collected between 14:30:00, September 4, 2019 to 23:00:00, September 9, 2019. The data were transferred to the server over LoRaWAN network using our own gateway as there was no commercial coverage at the site. The CO data from the reference Teledyne sensor were also collected over the same time period. All the sensors collected data at one-minute sampling interval. However, average reading for each ten minutes has been used for further investigation as the reference sensor provides calibrated output at that sampling interval. Thus a total of 772 samples are used for each sensor data

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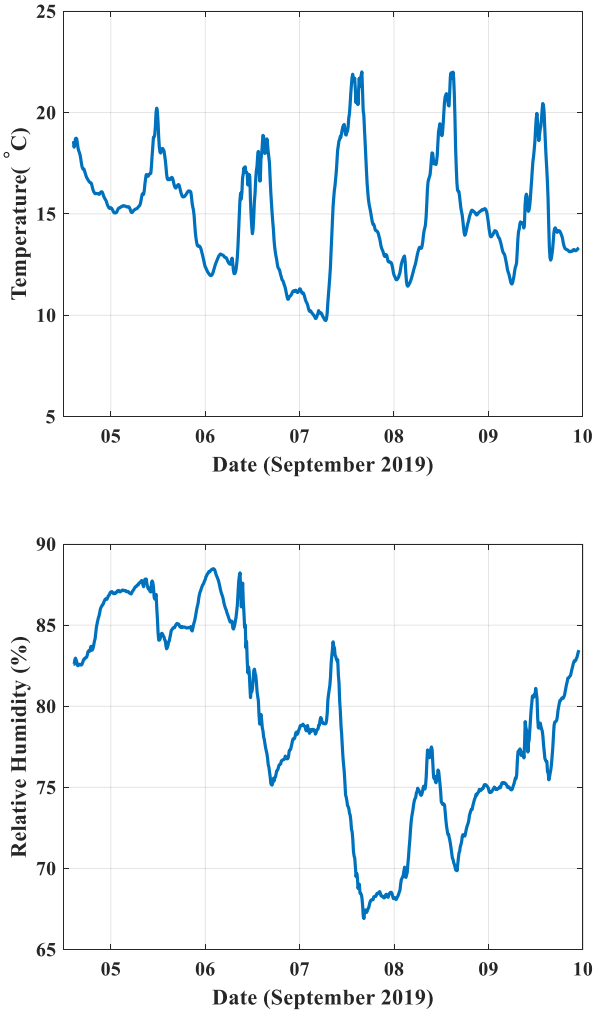


Fig. 11. Sensor S2 temperature and RH data.

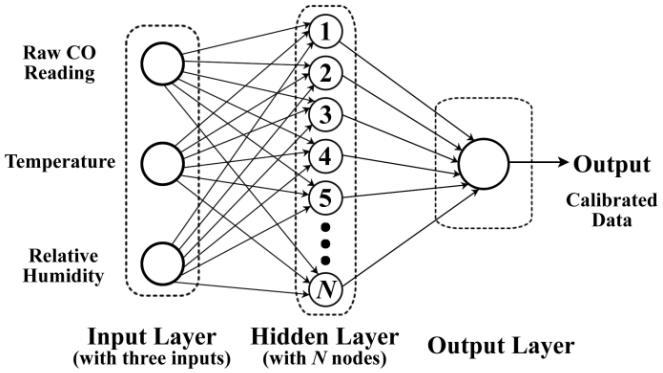


Fig. 12. Feedforward part of the ANN.

set. Fig. 9 shows the reference CO readings and our sensors' (S1, S2, and S3) raw data readings, respectively.

The CO sensors of the three nodes can be calibrated by comparing the sensors' data with the reading from the reference sensor. While the CO sensors come with factory offset and gain calibration, further correction is essential for mapping the raw measurements to the target pollutant concentration [4].

The sensor data were calibrated using ordinary least squares (OLS) regression by minimizing the sum of square differences

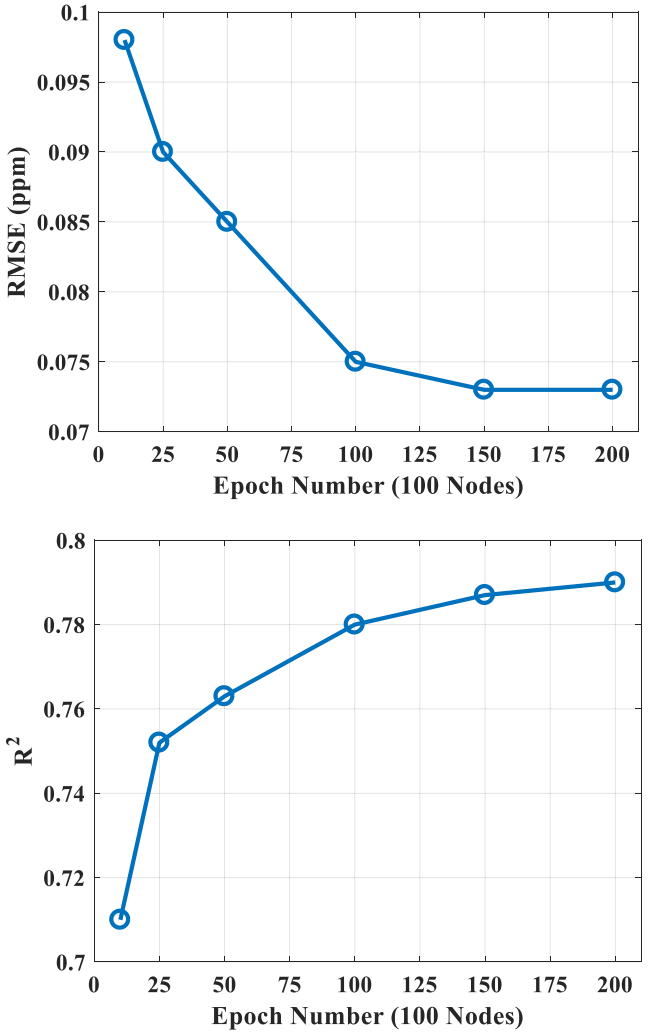


Fig. 13. Impact of epoch number on RMSE and R^2 for ANN calibration of sensor S1.

with the reference [2], [55]. Regression provided the intercept (offset) and slope (gain) values for each of the low-cost CO sensor. These values are used to determine the calibrated response by offset and gain calibration. Mean absolute percentage error (MAPE), coefficient of determination (R^2), and root-mean-square error (RMSE) have been calculated for the calibrated values with respect to the reference values (please see Table III). Fig. 10 (row 1) shows the calibrated sensor responses for OLS models for all three sensors.

Sensor response or sensitivity of the sensor can be affected by changes in the ambient temperature and RH requiring correction for these factors [56]. Multiple linear regression (MLR) can be used to approximate a linear combination of raw air pollutant concentration, temperature, and humidity measurements that best fits the target reference concentration for low-cost sensors [53], [57]–[59]. The developed sensors produce temperature and RH data (see Fig. 11) along with raw gas sensor responses and all these data can be used to perform temperature and humidity correction in the sensor response using MLR as shown in the following equation:

$$\hat{Y} = \beta_0 + \sum_{i=1}^{3} \beta_i X_i. \tag{1}$$

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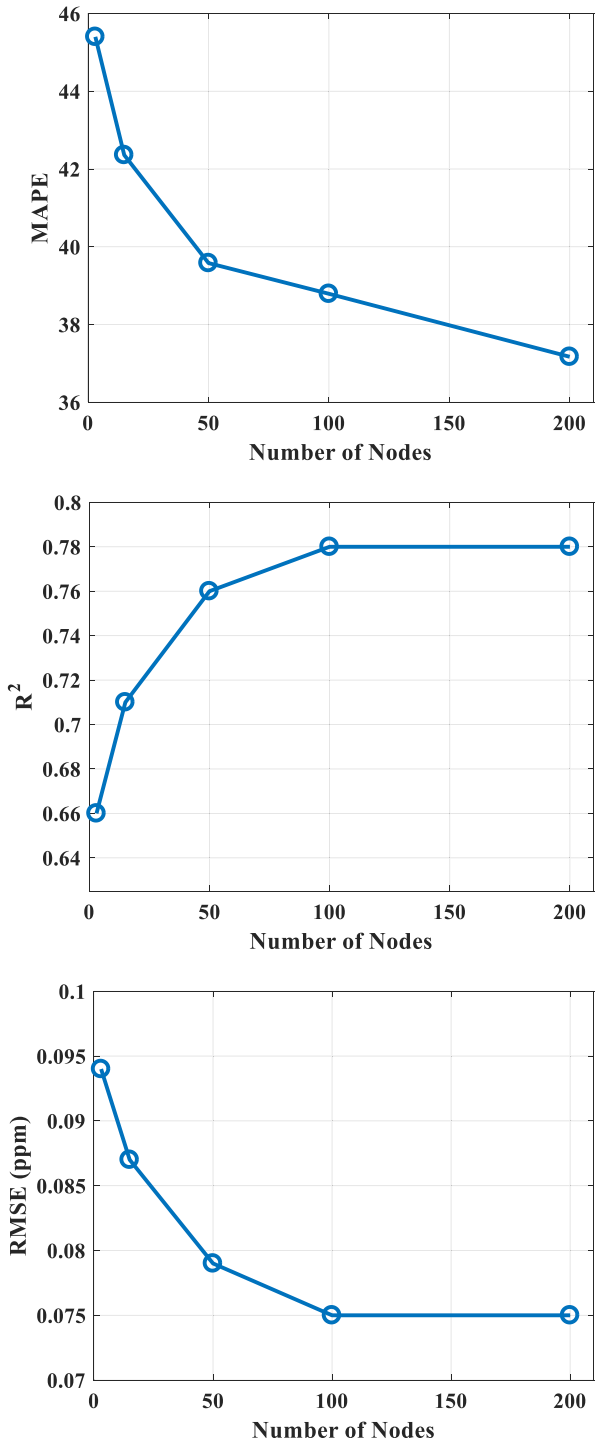


Fig. 14. Impact of number of nodes in the hidden layer of ANN on the calibration of sensor S1.

Here, $\hat{Y}$ is the calibrated CO reading, X represents the raw CO, temperature, and humidity readings obtained from the sensor node and β are the regression coefficients that are estimated by minimizing the mean square error (MSE) with the reference. Results show that the multivariate calibration using MLR can improve the calibration compared to OLS by considering temperature and humidity correction (please see Table III). Fig. 10 (row 2) shows the output of all three sensors after performing MLR-based calibration.

The application of more sophisticated machine learning-based techniques [60] has started to gain traction for calibrating low-cost sensors. Machine learning approaches have been successful in achieving multivariate calibration with NNs being one of the most popular methods for such calibration [50], [51], [55], [60]–[63].

Artificial neural network (ANN) was therefore explored to ascertain its suitability for calibrating the developed sensors. Raw CO sensor response, temperature, and RH from the developed sensors are used as ANN inputs, while the reference sensor data are used as the target output. The ANN is a fully connected multilayer perceptron (MLP). A shallow structure with a single hidden layer was chosen as extensive testing did not show any noticeable improvement in calibration performance with deeper structure consisting of multiple hidden layers. This also reduced the complexity and the training time. Fig. 12 shows the feedforward part of the ANN. The operation of the ANN can be explained with the aid of the following equations:

$$O_j = \varphi_{\mathrm{H}}\left[\sum_{i=1}^{3} w_{ij} X_i + b_j^{\mathrm{H}}\right] \tag{2}$$

$$\hat{Y} = \varphi_{\mathrm{O}}\left[\sum_{j=1}^{N} w_j O_j + b^{\mathrm{O}}\right]. \tag{3}$$

Here, X_i represent the three inputs obtained from the sensor node readings, O_j is the output of the jth node of the hidden layer, w_{ij} are the weights, b_j^{H} are the biases, and φ_{H} is the activation function for the hidden layer. $\hat{Y}$ is the output of the ANN, i.e., the calibrated CO readings, w_j represent the weights, b^{O} is the bias, and φ_{O} is the activation function of the output layer. After extensive testing, rectified linear unit (ReLU) was selected as the activation functions for both layers. The weights and the biases are adjusted during the training by backpropagation to minimize the MSE between the output, i.e., the calibrated CO reading and the ground-truth, the CO reading from the reference sensor. The ANN was implemented using Keras, the open-source NN library of Python.

Given the relatively small size of the data set, leave-one-out cross-validation method [64] has been used. The ANN model is trained with 771 input data and tested for the remaining data and the process is continued for every data sample. The training is performed for 200 epochs. It was observed that the rate of decrease for RMSE and the rate of increase of R^2 slow down around 100 epochs (please see Fig. 13 for an example of S1 training). Similar trends were also observed in the rate of decrease of MAPE. More epochs incur higher computational cost for the training and the small performance improvements do not justify going beyond 100 epochs. There is also the risk of overtraining resulting in poor generalization capability [65]. In a real-world setup with a large amount of training data, the number of epochs along with other hyperparameters will need to be carefully selected to reduce the error while not overtraining the NN. Rather than stopping the training at a fixed epoch number, more sophisticated strategies can also be

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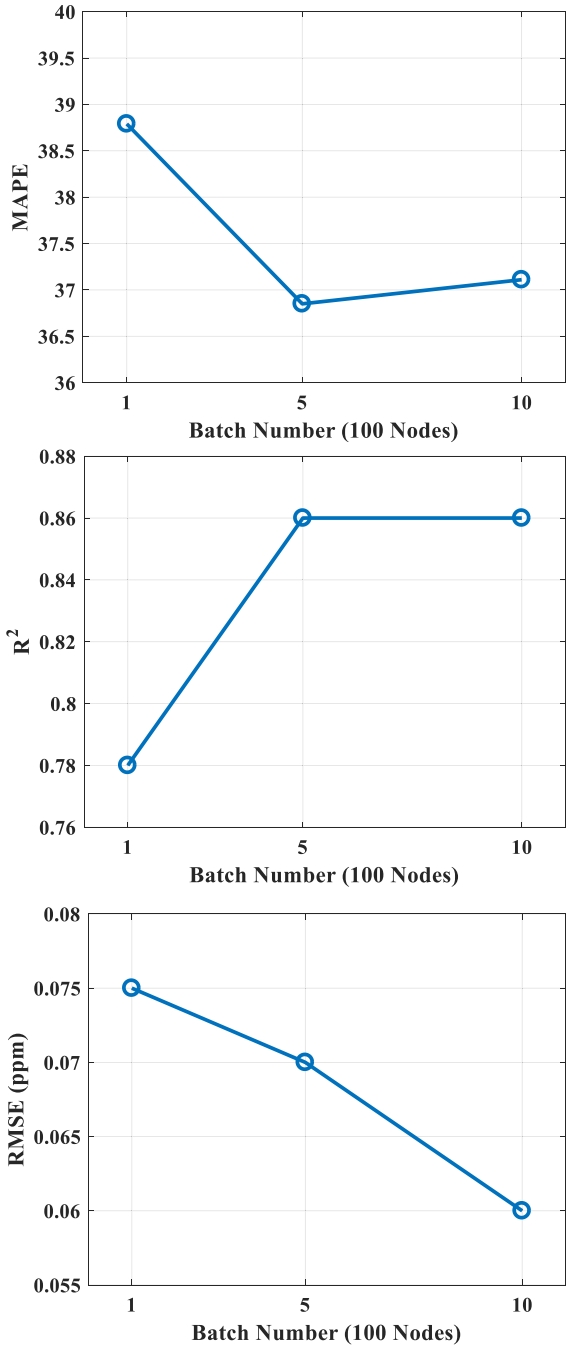


Fig. 15. Impact of batch size of ANN training on the calibration of sensor S1.

applied to retain the generalization capability of NN without overtraining [66].

Fig. 10 (row 3) shows the responses for all three sensors after ANN-based calibration with 100 nodes in the hidden layer with batch number set at 1. The key results are summarized in Table III. As can be observed, ANN shows significant improvement with respect to OLS and MLR methods. The ANN considers the nonlinear relationships between the input and output data along with offset and gain calibration and temperature and humidity correction.

Fig. 14 shows that the impact of number of nodes in the hidden layer on the calibration. As can be seen, there is diminishing return beyond 100 nodes and therefore the hidden layer was constructed with 100 nodes. Further improvements can be achieved by increasing the batch size (see Fig. 15) at the cost of more memory and longer training time.

For future investigations, time of the reading can be utilized as an input of the multivariate calibration. There is a clear temporal pattern in the CO reading with the dips occurring during the evenings and higher readings observed during daytime that coincides with increased human activities (e.g., traffic). Recurrent neural network (RNN) [62] can also be explored to exploit this temporal dependence.

IV. Conclusion & Future Works

This article reported the development of a functional LoRaWAN-based low-cost sensor node. The developed node can measure common ambient pollutants along with temperature and humidity. Three sensor nodes were manufactured and deployed for testing. The CO sensors were calibrated with the aid of a collocated accurate reference sensor using multiple calibration methods. ANN-based calibration method seems quite promising resulting in high R^2 values and low MAPE for ten-minute sampling interval. Therefore, the sensors calibrated by NNs-based method may be able to facilitate near real-time monitoring of air pollution over a large area.

The NO_2 and PM sensors also need to be calibrated by collocating them with reference sensors. Once all the calibrations have been performed, the sensors need to be redeployed for further testing and collecting data over a period of many weeks. The data collected can then be used to calibrate the sensors for long-term sensor drift and develop pollution forecasting models. While preliminary analysis was performed, the solar panel, power consumption, and the charging have not been tested extensively and will be a focus area for future research.

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